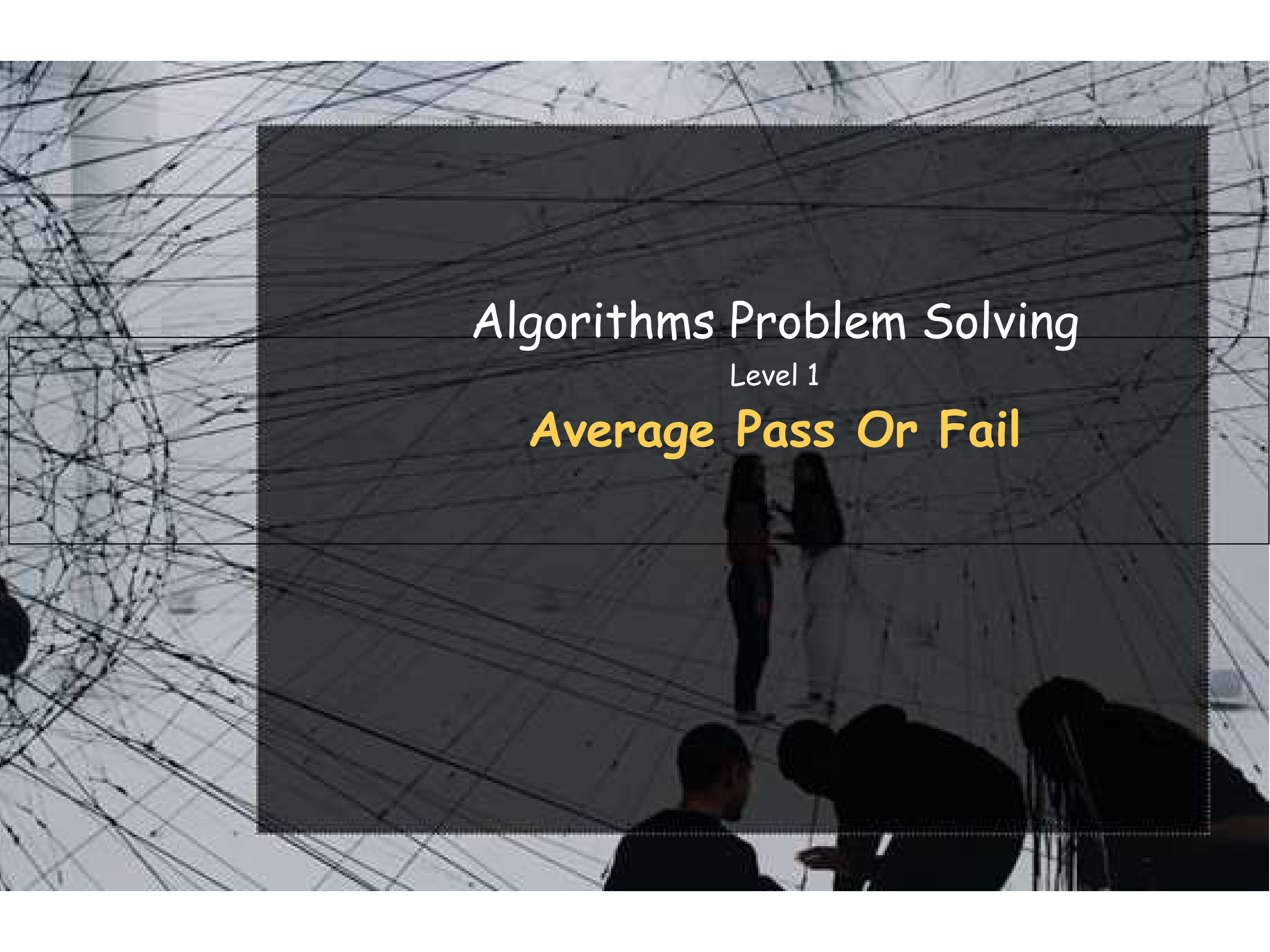


Algorithms Problem Solving Level 1

Average Pass Or Fail

The background of the slide is a grayscale photograph of a basketball court. In the center, two players are standing and facing each other. In the foreground, the silhouettes of several other players are visible, some looking towards the center. The court floor has a visible wood grain pattern.

Algorithms Problem Solving

Level 1

Average Pass Or Fail

Definition

- ◆ This program reads three marks from the user, calculates their sum and average, and checks whether the student has passed or failed.
- ◆ The student passes if the average is greater than or equal to 50.
- ◆ The program uses a structure, functions, floating-point division, and a Boolean result.

Explanation

- ◆ The struct `stMarks` groups the three marks:
 - `mark1` stores the first mark.
 - `mark2` stores the second mark.
 - `mark3` stores the third mark.
- ◆ `ReadMarks()` reads the three marks and returns an `stMarks` object.
- ◆ `sumMarks()` calculates and returns the sum of the three marks.
- ◆ `avgMarks()` calls `sumMarks()` and divides the result by 3.
- ◆ `(float)` is used to preserve the decimal part of the average.
- ◆ `checkPassFail()` checks whether the average is greater than or equal to 50 and returns true or false.
- ◆ `PrintResult()` displays the sum, average, and final PASS or FAIL result.
- ◆ In `main()`, the structure returned by `ReadMarks()` is passed directly to `PrintResult()`:
`PrintResult(ReadMarks());`

Steps in main()

1. Create the stMarks structure.
2. Read the three marks from the user.
3. Return the structure containing the marks.
4. Pass the structure to PrintResult().
5. Calculate the sum using sumMarks().
6. Calculate the average using avgMarks().
7. Check whether the average is greater than or equal to 50.
8. Return true for PASS or false for FAIL.
9. Print the sum, average, and final result.

💡 Example

◆ If the user enters:

Mark 1: 60

Mark 2: 70

Mark 3: 50

◆ The sum is:

$$60 + 70 + 50 = 180$$

◆ The average is:

$$180 / 3 = 60$$

◆ Since $60 \geq 50$, the student passes.

➡ Result

The sum of the three marks is: 180

The average of the three marks is: 60

You have PASSED

Note

- ◆ `checkPassFail()` returns a bool value.
- ◆ `true` means the average is 50 or more.
- ◆ `false` means the average is below 50.
- ◆ The condition `>= 50` means that an average of exactly 50 is considered a passing result.

⊘ Warning

- ◆ Do not confuse the average with the sum.
- ◆ The pass or fail decision is based on the average, not the total sum.
- ◆ Floating-point division is used so that decimal averages are preserved.



Comparison

PASS

- ✓ Average is greater than or equal to 50.
- ✓ checkPassFail() returns true.
- ✓ The program prints "You have PASSED".

FAIL

- ✓ Average is below 50.
- ✓ checkPassFail() returns false.
- ✓ The program prints "You have FAILED".

✦✦ Summary

- ✓ A structure can group multiple related marks into one object.
- ✓ The pass or fail decision is based on the average of the three marks.
- ✓ Functions can be reused by other functions to perform calculations and make decisions.

Thank you 😊